

Muhammad Zayan

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EDUCATION

Ghulam Ishaq Khan Institute (GIKI)

Bachelor of Science in Computer Science

- CGPA: 3.17/4.00

Topi, Pakistan

Sep. 2023 – May 2027

EXPERIENCE

CDIS Intern (Cloud, Digital Infrastructure & Security)

Systems Limited

June 2026 – August 2026

Islamabad, Pakistan (On-site)

- Architected an IDE-style RAG platform for IBM CP4I/ACE using FastAPI, React/TypeScript, and PostgreSQL (pgvector), integrating hybrid semantic/lexical search, HyDE, reranking, and async ingestion for code/docs while evaluating retrieval via Precision@K, Recall@K, MRR, and nDCG.
- Transferred RAG prototype capabilities into team CP4I/ACE Agent across 12 commits, engineering a 2-stage Node.js-to-ACE migration pipeline, deterministic ESQl/message-flow validation, token-aware rolling memory, and a human-gated self-learning lifecycle verified by 70 passing automated tests.
- Built an autonomous Hermes-based DevOps assistant featuring a stateful agentic tool-calling loop, extensible JSON-schema registry, and 9 non-destructive inspection tools for filesystem, GitHub, and Docker container observability with mocked integration tests.
- Deployed multi-service AI stacks on Red Hat OpenShift, designed Jira-governed release approval flows for Bitbucket/GitHub Actions CI/CD pipelines, and authored a 30-page architecture & operations manual containing 19 technical diagrams across 4 ValueOps repositories.

Machine Learning Intern | [GitHub](#) ↗

FlyRank AI

July 2026 – August 2026

Remote

- Designed and implemented an ML content opportunity and decay prioritization engine using Random Forest models, evaluating performance with Precision@K, ROC-AUC, and PR-AUC ranking metrics.
- Implemented GroupKFold cross-validation and client-grouped holdout splits to prevent target leakage, achieving a Precision@50 of 0.68 on client holdouts compared to a 0.24 baseline.
- Architected SIRA (Search Intelligence & Research Agent), an autonomous AI agent running Plan-Act-Observe-Evaluate loops to verify repository artifacts, audit feature safety, and inspect model metrics.
- Deployed a public GitHub Pages research paper with public-safety data checks, housing reproducible Jupyter notebooks, error analysis, and actionable Content Action Playbooks.

AI Intern

Integration Xperts

June 2025 – July 2025

Karachi, Pakistan (On-site)

- Engineered computer vision solutions using YOLO and MediaPipe for object detection and pose estimation.
- Optimized ML model inference times for real-time applications within cross-functional teams.
- Preprocessed datasets using Pandas and NumPy to streamline the training of deep learning models.
- Curated and annotated custom image datasets utilizing Roboflow to train robust object detection models.

PROJECTS

VisionForge | PyTorch, FastAPI, SigLIP 768D, YOLO11, Grad-CAM, Next.js 16

[Source Code](#) ↗

- Architected an open-source Computer Vision Workbench using FastAPI, PyTorch, and Next.js 16 (App Router), unifying inference, evaluation, and explainability across 247+ automated tests.
- Engineered a data quality pipeline implementing SHA-256 payload hashing and 768D SigLIP cosine similarity (≥ 0.92) to eliminate train-test leakage across dataset splits.

- Developed a PyTorch Grad-CAM explainability module to generate spatial feature heatmaps and quantify target bounding-box concentration for evaluating model feature reliance.
- Designed an 8-stage finite state machine lineage DAG engine using JSON-LD cryptographic hashes and Bessel-corrected statistics for experiment reproducibility.

VisionTraceAI | *React, FastAPI, Kafka, YOLO11, SigLIP, LangGraph*

[Source Code](#) ↗

- Architected a real-time, event-driven video analytics platform utilizing YOLO11, Kafka, and FastReID for cross-camera tracking.
- Engineered a zero-shot semantic search pipeline using Google's SigLIP and Qdrant for natural language text-to-video retrieval.
- Developed a React and FastAPI full-stack dashboard featuring a LangGraph autonomous agent to investigate spatial-temporal memory.

SentinelOps | *YOLO11s, FastAPI, DVC, MLflow, Evidently AI, PostgreSQL, Redis*

[Source Code](#) ↗

- Architected an MLOps industrial safety pipeline with fine-tuned YOLO11s, FastAPI, PostgreSQL, Redis, and Next.js, achieving **96.4% mAP@50** and **93.3% Precision** on PPE inspection datasets.
- Implemented automated dataset versioning and experiment tracking using DVC and MLflow/DagsHub to log hyperparameters, loss curves, and model artifacts.
- Integrated Evidently AI monitoring to execute continuous feature distribution drift detection and accuracy decay auditing on live video streams.

AI Gym Vision — Team Project @ Integration Xperts | *Python, MediaPipe, OpenCV*

[Source Code](#) ↗

- Developed a CV fitness assistant to track human pose in real-time utilizing the MediaPipe framework.
- Implemented vector mathematics to calculate joint angles for automated exercise form feedback.
- Collaborated in a team environment to integrate vision models with a user-friendly interface.

Matchmaker & Classification Engine | *Python, Scikit-Learn, Pandas, TF-IDF*

[Source Code](#) ↗

- Architected an IPO recommendation engine using TF-IDF vectorization and Cosine Similarity to map user skill profiles to career trajectories.
- Built a K-Nearest Neighbors supervised learning model to accurately classify structured datasets.
- Engineered a deterministic, rule-based chatbot with $O(1)$ dictionary logic and sanitization pipelines.

RELEVANT COURSEWORK

Core CS: Data Structures & Algorithms, Design and Analysis of Algorithms, OOP, DBMS, OS

Math & Stats: Linear Algebra, Discrete Mathematics, Inferential Statistics & Applied Probability

Specialized Electives: Machine Learning, Flutter App Development, DevOps

ADDITIONAL TRAINING

Stanford CS231n: Deep Learning for Computer Vision

TECHNICAL SKILLS

Languages: Python, C++, C, TypeScript, Dart, SQL, Bash/Shell Scripting

ML & Agent Frameworks: PyTorch, YOLO, MediaPipe, OpenCV, Scikit-learn, LangGraph, Model Context Protocol (MCP), pgvector, Qdrant

Tools & Infrastructure: Git, GitHub, GitHub Actions, CI/CD, Docker, Red Hat OpenShift, Kubernetes, DVC, MLflow, AWS, Linux, PostgreSQL, Redis